Prof. Tashfeen
CS 2163: Java

Assignment 1

The following questions are meant to be done in order since they build on each other.

Question 1. Read the relevant parts of the textbook as described in the online classroom.

Question 2. Download and install the latest Java Platform, Standard Edition Development Kit (JDK). You might need to include JDK's path with your system's path, if you don't know what this means or how to do it, please see the professor in his office hours or visit the third floor of the library.

Question 3. Watch either both or one of the following,

- How to Run Java Programs With Command Prompt (cmd) and Notepad in Windows
- Compile and Run a Java Program Using Command Line Terminal on Mac

These videos will familiarise you with how to write a program in a text editor and run it using the command line/prompt. You'll be doing this all semester so learning this now will save you a lot time. If you use GNU/Linux, I assume you already know what you're doing.

I recommend Sublime Text as your first text editor (though you can use any). Note that technically it is not free but it has the "unlimited free trial" philosophy, i. e., you can legally keep using it until you're a 100 and then if you feel like you have had a good use out of it, you can finally purchase it. Or, not.

Question 4. Write a Java program that generates a random integer between 0 and 9 and then asks the user to guess it. You may start by modifying the code in listing 1. Please substitute your name in-place of mine.

```java
/**
 * Name: Tashfeen, Ahmad
 * File: TashfeenAhmad.java
 * Description: Random number guessing game.
 */
public class TashfeenAhmad { // substitute your name inplace of mine
    public static void main(String[] args) {
        int lower = 0;
        int upper = 9+1;  // exclusive
        int randomInt = (int) ((upper - lower) * Math.random() + lower);
        String formatString  = "Random number between %d and %d: %d";
        System.out.printf(formatString, lower, upper-1, randomInt);
    }
}
```

Listing 1. A complete Java program to print a random integer between 0 and 9

Save the listing 1 code in a file called `TashfeenAhmad.java` (again, this is just an example, use your name) and run,

```
javac TashfeenAhmad.java
java TashfeenAhmad
```

Requirements

Note that *these are not merely suggestions*. You will lose points (possibly all of them) if you don't fulfil these. Even if your program works.

1) Your program must compile and run from the command line using `javac` and `java` commands.
2) Your program should prompt the user to guess an integer between the specified range.
3) If the user inputs an integer outside of the range then your program should print a message, e. g., "Out of range." and exit gracefully.
4) Your program should print a success message if the user inputs a correct number, e. g., "Correct."

5) If the user guesses the wrong number then your program should tell the user that their guess is incorrect. It should also print out the number that would have been correct.
6) Your class file should be named `LastnameFirstname.java`, e. g., for me it is `TashfeenAhmad.java`. If you used the `TashfeenAhmad.java` code then don't forget to change the class name (and therefore the filename).
7) You must close your scanner.
8) If you've been asked to demonstrate your program to the professor then you'll need to do so in his office hours or make an appointment. If you've been asked to demo your program and you fail to do so, you will receive a zero in the assignment.

Hints

You'll need to import the scanner class and read in an integer. Make sure to close the scanner once you're done using it. You may then use branching (`if else` statements) to check for the correctness of the guess and similarly if it is in the specified range. You may use a combination of `System.out.println()` and `System.out.printf()` to display messages.

Example Executions

See figure 1 for example of executions.

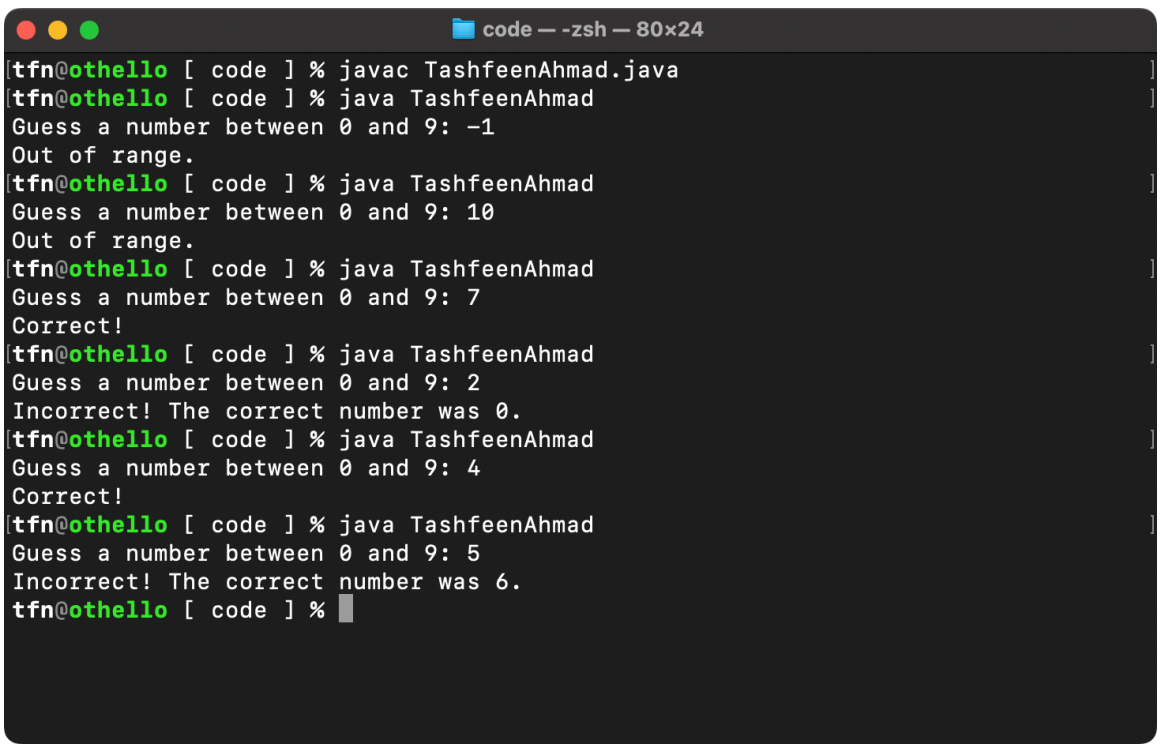

Figure 1. Example execution for the assignment 1 program.

Submission Instructions

1) Submit your `LastnameFirstname.java` file. Note that this is not the `LastnameFirstname.class` file.
2) Submit a screenshot like the one in figure 1. Check all input cases as shown. Keep running your program till you get at least a guess right and a wrong.

You will not receive any credit if any of the files to be submitted are missing.